Cheat Sheet

Scratch blocks (weeks 1–6) and Python turtle (weeks 7–12)

Scratch blocks

Events

Start the code. Message names must match!



Motion

n sides: turn $360 \div n$. The middle is (0, 0).



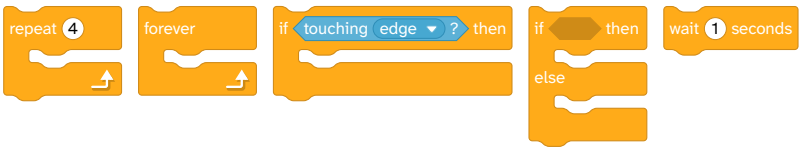
Looks & Pen

Costumes in a loop = animation. Pen: Extension button.



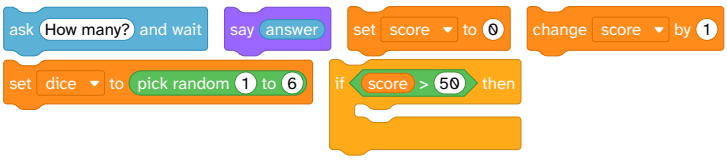
Control

Loops repeat what is inside. **if**: only when true. **else**: when false.



Ask, variables, random

The reply goes into **answer**. A variable is a labeled box.



My Blocks

Make a Block with a name and inputs, then use it.



Python turtle in Thonny

Getting started

`import turtle`	Load the turtle tools.
`t = turtle.Turtle()`	Make a turtle: `t`.
`screen = turtle.Screen()`	The window.
`turtle.done()`	Keep it open (last line).

Moving and drawing

`t.forward(100)`	Walk 100 steps.
`t.back(50)`	Walk backwards.
`t.left(90)`	Turn left 90 degrees.
`t.right(90)`	Turn right 90 degrees.
`t.penup()`	Pen up: move, no line.
`t.pendown()`	Pen down: draw again.
`t.goto(x, y)`	Jump to (x, y).
`t.setheading(90)`	Face up (0 = right).
`t.circle(50)`	Circle, radius 50.
`t.dot(20, YELLOW)`	A filled dot.
`t.write("Hi!")`	Write on the picture.

Colors and style

`t.color("#0072B2")`	Pen and fill color.
`t.color(BLACK, ORANGE)`	Lines, then fill.
`t.begin_fill()`	Start coloring in...
`t.end_fill()`	... fill it now!
`t.pensize(3)`	Thicker lines.
`t.speed(0)`	Fastest drawing.
`t.hideturtle()`	Hide the arrow.
`screen.bgcolor(BLACK)`	Background color.

Color-blind-safe colors

ORANGE	`#E69F00`	SKY_BLUE	`#56B4E9`
GREEN	`#009E73`	YELLOW	`#F0E442`
BLUE	`#0072B2`	VERMILLION	`#D55E00`
PURPLE	`#CC79A7`	BLACK	`#000000`

Thonny keys

Run F5 Stop Cmd+F2 Save Cmd+S Bigger text Cmd++

Python words

`print("Hello")`	Show text in the Shell.
`# a comment`	A note for humans.
`size = 50`	A variable (labeled box).
`answer = input("Size? ")`	Ask. The answer is text.
`n = int(answer)`	Text to a number.
`import random`	Load the surprises.
`random.randint(1, 6)`	Random: 1 to 6.

Loops, choices and functions

```
for i in range(4):   # repeat 4 times
    t.forward(100)   # inside the loop
    t.left(90)

if n > 10:
    print("Lots!")
elif n == 10:
    print("Exactly ten")
else:
    print("A few")

def square(size):    # make a command
    for i in range(4):
        t.forward(size)
        t.left(90)

square(50)           # use it!
```

Compare: > bigger, < smaller, == equal, != not equal.

Python is picky!

- Small letters: `print`, not `Print`.
- A colon : ends every `for`, `if`, `else`, `def` line.
- Lines inside are pushed in 4 spaces (Tab).
- Every (needs a); every " needs a partner.
- Never name a file `turtle.py` or `random.py`.